

ENERGY INTELLIGENCE • MICROSOFT FABRIC-NATIVE

EnergyLens

Turn every building into a self-diagnosing,
decarbonizing asset.

From raw sensor to audit-ready carbon report — EnergyLens sees, explains and acts on commercial-building energy, on one cloud-native platform with a built-in AI copilot.

9 analytics pages • web app • AI copilot



● WHY THIS MATTERS

The largest source of EU emissions is hiding in plain sight

40%

of EU energy is used by buildings

residential + non-residential

36%

of EU energy-related CO₂

the single largest sector

2030-

2033

EU performance deadlines

EPBD minimum standards, binding

Yet most portfolios still run on monthly bills and spreadsheets.

They are blind to the anomalies bleeding money every night, to a wall of tightening regulation — EPBD, EnEfG, GEG, CSRD, the EU Battery Regulation — and to the cheapest decarbonization wins. EnergyLens closes that gap for the mid-market the incumbents ignore.

One platform. Three layers. Every building, understood.

It unifies meters, sensors, weather, solar and battery data — then turns it into decisions, compliance and savings.

01

See

A 9-page Power BI report — consumption, anomalies, forecasting, sustainability, HVAC, real-time IoT and battery ROI — live from the Fabric Lakehouse.

02

Operate

A web app that wraps it all: secure multi-tenant access, building deep-dives, an EU-compliance hub, recommendations, alerts and one-click reports.

03

Ask & act

An AI copilot that answers in plain language by running real queries over your data — and can take action, not just describe it.

Plus a compliance brain — CRREM, EnEfG, GEG, CSRD/ESRS E1 and the EU Battery Regulation, scored per building.

● HOW IT'S BUILT

One copy of your data — from raw signal to audit-ready truth

Sources

Meters, bills, BMS sensors (BACnet/Modbus/MQTT), weather (Open-Meteo), prices (ENTSO-E), grid CO₂, solar & battery telemetry

Bronze

Raw, exactly as ingested

Silver

Cleaned, normalized, validated — one sourced reference layer for every factor & tariff

Gold

Business-ready KPIs, anomalies and simulations the report reads live

Three validated paths into Fabric: Power BI DirectLake · custom React via the SQL endpoint · the AI copilot via tool use.

PART 1 • SEE

The analytics report

Nine pages that turn raw energy data into decisions — live, on every building.

Nine pages, one living model

1 Portfolio

How the whole portfolio is doing this month

2 Building

What one building is doing right now

3 Anomalies

Where energy is wasted or a fault is forming

4 Forecast

What next month and quarter will cost

5 Decisions

What to do, in what order, and what it saves

6 Sustainability

Carbon footprint, EU report-readiness

7 HVAC

Heating & cooling efficiency, hour by hour

8 Real-time IoT

Live faults, diagnosed and priced

9 Battery ROI

Should you invest, and when it pays back

● **PORTFOLIO & BUILDING INTELLIGENCE · PAGES 1-2**

See your whole portfolio — then any building, live

EnergyLens ranks every building by energy intensity against its type, surfaces the worst performers, and lets you drop into a single building's 24-hour profile in one click — cost, peak and carbon, all climate-adjusted.

- **Fair comparison** — climate-adjusted EUI normalizes for weather, so no building is unfairly judged
- **Find the money** — worst-3 buildings and overnight 'base load' waste surface instantly
- **From portfolio to plug** — one click from the C-suite view to a facility manager's hourly curve

The whole portfolio, and any building, on one model.

[screenshot: portfolio + building]

drop your screen recording here

kWh/m²

the master metric

24h

hourly detail

Catch waste tonight; budget for next quarter

A single detection engine flags consumption spikes, COP drops, solar underperformance and after-hours waste — each with a probable cause and a euro figure. A per-building Prophet model then forecasts cost and carbon weeks ahead.

- **Diagnosis, not noise** — 7 rule families, trilingual, with an estimated € loss on every anomaly
- **Forecast that learns** — Prophet with weather and occupancy as drivers, per building
- **Plan with confidence** — forecast cost and CO₂ with an uncertainty band

Stop bleeding money now — and see the bill coming.

●
[screenshot: anomalies + forecast]

drop your screen recording here

7

anomaly types

90d

forecast horizon

A ranked to-do list that pays for itself

EnergyLens converts findings into ranked actions — each with annual savings, payback, effort and the exact German or EU grant that helps fund it. It's a multi-country compliance brain, not a generic dashboard.

- **Ranked by impact** — $\text{priority} = \text{saving} \times \text{urgency} \div \text{effort}$, so the best move is on top
- **Money attached** — €/kWh/CO₂ per year, payback in years, for every recommendation
- **Grants matched** — KfW, BAFA, EEG and YEKA programs matched automatically

Every insight becomes a funded, prioritized action.

●
[screenshot: recommendations]

drop your screen recording here

5

country regimes

€

savings + grants

Audit-ready carbon — the page that sells the platform

Scope 1, 2 and 3 emissions, an area-weighted EPC map, a CRREM stranding view and a 0–100 ESG score — mapped to ESRs E1 and built on traceable, sourced factors. Every disclosed kilogram traces back to a raw bill.

- **Regulator-grade lineage** — one sourced, year-indexed factor table — every kg CO₂ is traceable
- **Done right** — area-weighted EPC, dual-method Scope 2 — the things auditors check
- **Honest by design** — screening estimates are labelled as such — reporting support, not greenwashing

ESRS-E1-aligned reporting support, with traceable lineage.

●
[screenshot: Scope 1/2/3 + EPC]

drop your screen recording here

1/2/3

full GHG scopes

ESRS E1

aligned

Know when your heating is quietly failing

Heat-pump performance against its rating, supply/return temperature deltas and HVAC's share of load — compared within equal weather bands, so a normal cold-day dip is never mistaken for a fault.

- **Fair COP** — performance judged within equal outdoor-temperature bands
- **Spot the drift** — supply/return delta-T as a live efficiency signal
- **Right-size retrofits** — envelope and renovation priorities, building by building

Catch a failing heat pump before the bill does.

●
[screenshot: HVAC / COP]

drop your screen recording here

COP

actual vs rated

ΔT

efficiency proxy

Your building talks back — in real time

EnergyLens listens to every sensor across BACnet, Modbus and MQTT. The moment an air handler drifts or a chiller starts wasting energy, it doesn't just alarm — it diagnoses the fault, names the probable cause, and puts a euro figure on it.

- **Diagnosis, not alerts** — ASHRAE Guideline 36 logic — 12 rules with confidence and probable cause
- **Speaks every protocol** — BACnet, Modbus, MQTT and ~31 sensor types, normalized automatically
- **Counts the cost** — every fault carries a live euro estimate, so teams fix what matters first

Enterprise-grade fault detection — at mid-market reach.

[screenshot: live IoT page]

drop your screen recording here

12

diagnostic rules

~31

sensor types

Answer the CFO's question: is a battery worth it?

EnergyLens compares battery chemistries and dispatch strategies across countries, then returns payback, NPV and IRR — only ever using EU-compliant chemistries, with real, sourced electricity prices.

- **Real finance** — NPV with discounting, inflation and salvage; Newton-Raphson IRR; payback
- **Strategy-aware** — self-consumption, peak-shaving and arbitrage compared head-to-head
- **EU-compliant only** — scenarios respect the EU Battery Regulation 2023/1542

Investment-grade ROI, not a back-of-envelope guess.

●
[screenshot: battery ROI]

drop your screen recording here

NPV ·
IRR
real appraisal

12×7×8
scenarios

PART 2 · OPERATE

The web application

Where the locked report becomes a secure, multi-tenant product.

● THE PLATFORM LAYER

More than a dashboard — a product

A Next.js + FastAPI + PostgreSQL application wraps the report with secure access, custom Fabric-powered pages, one-click PDF reports and a full audit trail. Three layers of access mean every customer sees only their own buildings.

- **Secure by design** — row-level security on data, module gating on navigation, tiers on features
- **Built for teams** — org accounts, member invites, roles, Stripe self-serve billing
- **Resilient** — graceful degradation, audit logging, mobile-ready, accessible

The report stays a locked template; the app makes it a product.

●
[screenshot: portfolio dashboard]

drop your screen recording here

3 layers

of access control

RLS

data never crosses

● THE COMPLIANCE HUB

Stay ahead of the EU agenda — in one place

MEPS renovation risk, CRREM stranding years, flexibility readiness, EU-Taxonomy alignment and an ESRs-E1 summary — the regulatory questions that keep asset managers awake, answered per building, with an honest ‘indicative’ caveat.

- **Renovation radar** — EPBD 2030/2033 risk banding by EPC, building by building
- **Stranding foresight** — indicative CRREM 1.5°C pathways and the year each asset strands
- **Report-ready** — ESRs-E1 energy and Scope 1/2/3 summary, exportable to PDF

How we stand next to Schneider and Siemens on sustainability.

●
[screenshot: compliance hub]

drop your screen recording here

5

EU features

Schneider
Class, mid-market

● WORKFLOW · ACTIONS & ALERTS

From insight to action — and accountability

Recommendations and anomalies flow into a working queue: assign a status, acknowledge an alert, and every change is written to an audit log. Insights don't die in a dashboard — they get done.

- **A real queue** — status workflow on every recommendation; triage on every alert
- **Nothing slips** — 'unhandled' = unresolved and unacknowledged, surfaced in a nav badge
- **Accountable** — every change logged with who, when and from where

Insights become tracked, accountable work.

●
[screenshot: actions / alerts]

drop your screen recording here

37

live recommendations

Audit

every change

PART 3 • ASK & ACT

The AI copilot

Conversational analytics that runs real queries — and can act.

● AI COPILOT

Ask in plain language. It queries your data — and acts.

Most BI ‘copilots’ summarize a page as text. EnergyLens runs real queries over your Lakehouse through six production tools — and one of them writes back, so the copilot can change a status, not just describe it.

query_kpi	KPIs for a building
compare_buildings	rank / compare buildings
list_recommendations	recommendations by priority
get_anomalies	anomalies by severity / type
simulate_battery_scenario	payback / NPV / IRR
update_action_status	change status (a write)

●
[screenshot: copilot chat]

drop your screen recording here

Provider-flexible: Anthropic, Azure OpenAI or Mock — production is one setting. Security enforced on every call.

PART 4 • THE INNOVATION ENGINE

From dashboard to autonomous advisor

The working platform is the foundation. This is the 12-month build the EXIST grant funds.

Four moves from analytics to autonomy

A

Agentic AI copilot

Reasons over your Lakehouse with real tool use today — evolving into a copilot that proposes and sequences retrofit and dispatch plans.

P

Predictive intelligence

Beyond forecasting: ML-driven predictive maintenance that catches faults before they cost money.

O

Closed-loop optimization

From simulation to control — optimal battery dispatch, HVAC setpoints and automated demand response (EU DR Network Code).

M

Auditable carbon method

A regulator-grade, fully traceable carbon-lineage methodology — research we are formalizing with BSBI.

Live today: the full platform above. The innovation engine is our 12-month roadmap — exactly what we are seeking the EXIST grant to fund.

What the EXIST grant turns into a company

Months 1-3

Foundation & first pilot

BSBI campus pilot live; predictive-maintenance models in training; methodology paper outlined with BSBI.

Months 4-6

Agentic copilot

Copilot moves from answering to proposing & sequencing retrofit and dispatch plans; 2-3 paying pilots.

Months 7-9

Closed-loop control

Optimization for battery + HVAC + demand response; first paying customers; UG incorporation.

Months 10-12

Scale-ready

Auditable-carbon method published; 5+ customers, €5-10k MRR; Microsoft Partner track.

PART 5 • THE OPPORTUNITY

Market & strategy

A large, regulation-driven market — and a near-empty niche to own.

● MARKET SIZE

Big, growing, and pulled forward by regulation

€18B

TAM — EU commercial buildings

~5M buildings × €100–700/mo

€4.3B

SAM — DACH focus

highest-value market

\$8.4→2

4B BEMS software market

2026 → 2035

8.7%

BEMS market CAGR

through 2035

Five tailwinds, all at once

Binding EU climate regulation (2024–2033) • the post-2022 price shock made energy a board topic • Microsoft Fabric matured • IoT sensor costs collapsed • AI made pattern recognition affordable. The reasons to buy are structural and accelerating.

A near-empty intersection: Fabric-native + EU niche + mid-market

PLAYER	WHERE THEY ARE	THE GAP WE FILL
Measurabl / Aquicore	US enterprise, well-funded	mid-market EU coverage
Cortexa	EU enterprise	mid-market access & price
Schneider EcoStruxure	peer: auditable reporting + AI	multi-protocol IoT + Fabric-native
Siemens Designo	BMS / building automation	the analytics layer, not a BMS

Our angle: an AI copilot that reasons over the Lakehouse via tool use — from under €299/building/month.

● PRICING & UNIT ECONOMICS

SaaS economics that work for a lean team

Insight

€99

/building/month

SME offices

Monitor

€299

/building/month

mid-market property mgmt

Copilot

€699+

/building/month

enterprise / IoT-equipped

Custom

€5-50k

/building/month

10+ building portfolios

36-108×

LTV / CAC

2-4 mo

payback period

~85%

gross margin

First three months free (pilot) to win reference cases • break-even at customer #2.

Bottom-up, validation-first — then expand

Now → 6 mo

Free university & DACH mid-market pilots that close in 2–8 weeks. BSBI campus pilot first.

6–18 mo

Convert pilots to paying customers; case studies open hotels, healthcare and REITs.

18 mo+

Microsoft Partner Network co-selling; expand DACH → BeNeLux → Southern Europe.

The 3-year vision: the default Fabric-native energy-intelligence platform for European commercial real estate — 100+ buildings, €1–2M ARR, a Microsoft partner in climate-tech.

● THE FOUNDER

Energy domain depth × modern data engineering — a rare pairing

5 weeks

to build the whole stack, solo

pipeline → app → AI copilot

DP-600

Microsoft Fabric certified

few solo founders hold it

B.Sc.

+M.Sc.

Energy Engineering + Management

Yaşar + BSBI (June 2026)

Most BI developers don't know kWh/m² or Scope 2; most energy engineers don't know medallion architecture or LLM tool use. EnergyLens sits at that intersection — built end-to-end, solo, accelerated by AI-assisted development, with an 18-month visa runway to incorporate in Germany.

THE ASK

How you can help

1 · EXIST academic mentor

EXIST is the German federal founder grant (~€65k / 12 months); it requires an academic mentor letter. Since EnergyLens grew from my M.Sc. work, I'd be honored if you'd serve as mentor — one short letter (I'll draft it) and a couple of alignment chats.

2 · BSBI campus pilot

A free, six-month, secure pilot I run end-to-end — real-time visibility for the facilities team, a named case study for BSBI, and validation for my capstone. No cost, no commitment.

3 · Ongoing advisory (optional)

Occasional monthly / quarterly check-ins on the energy and academic angles — no formal commitment, just openness to input.

Thank you.

I'd welcome your guidance — and your support on the EXIST application.

Ali Mert Özdemir · energylens.eu · Microsoft DP-600

Demo: <https://youtu.be/0liqssHYtkY>

